Supporting Information for "Regional Heterogeneity in Lunar Deep Regolith Conductivity from a Per-Site Retrieval Against the Restored Apollo 15 and 17 Heat-Flow Experiment Record"

A Self-Contained Walk-Through of the Methodology

R. P. Gregorio

May 8, 2026

Contents

How to read this document

This Supporting Information (SI) is written as a self-contained tutorial. Each chapter starts from first principles and assumes only an undergraduate background in differential equations, basic statistics, and rudimentary Python. If you are an experienced lunar thermophysics modeller, you may skim Chapters 2–5 and jump to Chapter 6. If you are a planetary-science student or non-specialist reader opening these methods for the first time, please start with Chapter 1 (a non-technical narrative of the entire study) before reading the rest in order.

Visual conventions (used throughout):

Aside

Coloured *aside* boxes contain pedagogical context, intuition, or historical notes that the main text does not strictly require but which help build understanding.

Key result

Green *key result* boxes contain a takeaway you should remember after reading the surrounding section.

Pitfall

Red *pitfall* boxes flag a place where it is easy to make a subtle mistake; we describe the mistake and the way to avoid it.

Chapter 1

What this study did, in plain English

This chapter is for the reader who wants to understand what was done, why it matters, and how each piece fits together — without having to follow any equations. Every technical concept introduced here is unpacked in a later chapter, and a forward-pointer ("§ XX.X") tells you where to look. If something here is unclear, the right move is to read this chapter once for the narrative and then dip into the later chapters as the equations become relevant.

1.1 The setting

The Moon has no atmosphere and no oceans, so its surface heats and cools by hundreds of kelvin every ∼29.5-Earth-day lunar day. The thin, broken-up layer of rock at the surface — the *regolith* — is what insulates the deeper interior from those extreme swings. How effectively it insulates is determined mostly by a single material property, the *thermal conductivity* (the ability to carry heat through the material). Most spacecraft analyses of the Moon, including calculations of how stable buried water-ice is near the poles and how hot a future lander's payload will get, treat this conductivity as a single fixed number that applies everywhere on the Moon. That assumption has never been tested directly, because nobody has ever measured the conductivity of the deep regolith *in place* — only inferred it from how the surface looks in infrared images from orbit.

1.2 The data we used

In 1971 and 1972, the Apollo 15 and Apollo 17 astronauts buried two arrays of thermometers ∼2.4–2.5 metres deep into the lunar surface. These thermometers ran almost continuously until 1977 and recorded the only *in-situ* measurements of subsurface temperature ever made on the Moon. The original data tapes were lost for decades. Nagihara et al. (2018) restored the full 1971–1977 record, and that restored record is the dataset we use here. The detailed instrument description is in §2.

1.3 The model we built

We built a one-dimensional, time-dependent computer model of the heat flow through a vertical column of regolith. At each instant the model takes the sunlight falling on the surface (computed from exact spacecraft-grade ephemerides; §4) and asks: given the regolith's thermal properties, what temperature does each depth settle to over many lunar days? Solving that question requires a numerical scheme called Crank–Nicolson (§5); think of it as a careful bookkeeping algorithm that updates every depth in the column by a small time step, conserves energy at each step, and is stable enough to keep running for hundreds of simulated lunar days without drifting.

The thermal properties — conductivity, density, and heat capacity — come from a published global model by Hayne et al. (2017) that was fit to infrared brightness measurements taken from orbit. We use their model exactly as published, except that we treat one number (the deep conductivity, called K_d) as adjustable rather than fixed.

1.4 The single experiment

The experiment is a one-parameter fit. We sweep the deep conductivity K_d across a range of plausible values, run the full heat-flow model for each value, and ask: which K_d makes the model's predicted deep temperatures best match the Apollo thermometer readings? We do this independently for each landing site. The full procedure is in §6.

There is one subtlety. The shallow thermometers (above ∼80 cm depth) are known to be contaminated: they are mounted inside fibreglass tubes that conduct heat down from the *surface*, where the temperature swings are huge, and so they read warmer than the regolith would predict on its own. We exclude those shallow readings from our fit. Crucially, we identify which sensors to exclude using a *model prediction* — specifically, the predicted day–night swing as a function of depth — rather than by looking at where our model fits the data poorly. This avoids the trap of throwing away exactly the data points the model fails on.

1.5 What we found

When we run the experiment, the two Apollo sites give very different answers:

- Apollo 15 prefers $K_d \approx 4.85$ mW/(m K).
- Apollo 17 prefers $K_d \approx 11.23$ mW/(m K).

The published global value is 3.4 mW/(m K). Both sites *disagree* with the published value, and the two sites disagree *with each other* by more than a factor of two. The disagreement is much larger than the uncertainty on either measurement (quantitatively, ∼7.5 standard deviations on a leave-one-out jackknife; §6).

1.6 Why that matters

Two things follow from this result.

First, the thermal conductivity of the deep lunar regolith is regionally heterogeneous — it is genuinely different at different places on the Moon, even at non-polar latitudes where the regolith is comparatively well-studied. Whether this reflects differences in mineralogy, in compaction with depth, in particle-size distribution, or in some other property that varies regionally, is a question for follow-up work. But the assumption that one global value of K_d describes the whole Moon is not consistent with the in-situ data.

Second, anyone using the published global value to compute the cold-trap depth where polar water-ice is stable, or to set the heat budget for a lander instrument, is implicitly accepting an uncertainty at least as large as the inter-site disagreement we report. That uncertainty is large enough to matter for several mission-design questions, and it should be reflected in the error bars of any analysis that propagates K_d through a temperature calculation.

1.7 What this study cannot tell you

There is one important caveat. The two unknowns at the bottom of any buried-thermometer column are the deep conductivity K_d and the heat coming up from the deep interior, Q_b. At long times the two unknowns enter the equilibrium temperature gradient *only as their ratio* Q_b/K_d, so any single buried column cannot tell them apart on its own. We adopt the canonical published Q_b

values for each site (Langseth et al. 1976; Nagihara et al. 2018), and the absolute K_d we report inherits whatever uncertainty those values carry — about ±30% at A15.

The good news is that the *difference* between A15 and A17 is not affected by this caveat: any global rescaling of Q_b rescales both sites equally, leaving the contrast fixed. So the central finding (regional heterogeneity) is robust to the Q_b uncertainty; the absolute A17 number is not. We discuss this in detail in §6.

1.8 Where to read next

The rest of this document fills in each piece of the narrative above:

- Chapter 2 — the lunar thermal environment and how the Apollo HFE instrument actually worked.
- Chapter 3 — the heat equation, derived from first principles.
- Chapter 4 — the Hayne et al. (2017) regolith property functions, and the SPICE/DE440 surface-forcing chain that drives the model.
- Chapter 5 — the Crank–Nicolson numerical solver and its convergence diagnostics.
- Chapter 6 — the per-site K_d retrieval, including the leave-one-out jackknife uncertainty, the borestem-exclusion diagnostic, and the K_d/Q_b degeneracy.
- Chapter 7 — independent cross-checks of the solver.
- Chapter 8 — a reproducibility guide that walks through the source code file by file.
- Chapter 10 — a glossary of terms and symbols.

Take-away

The study uses Apollo-era buried thermometers to measure the *insulation strength* of the deep lunar regolith independently at two landing sites; the two sites disagree by more than a factor of two; therefore the regolith is regionally heterogeneous, and the single global value used in current spacecraft analyses understates the true uncertainty.

Chapter 2

The Moon as a Thermal System

2.1 Why the Moon's surface temperature varies so much

The Moon has no atmosphere and no oceans. The only things buffering its surface temperature against the day–night solar cycle are the *thermal inertia* of the regolith (the loose, broken-up rock covering the surface) and, to a lesser extent, radiative re-emission to space. Thermal inertia is conventionally defined as

$$I = \sqrt{K \rho c_p}, \qquad [\,\mathrm{J\,m^{-2}\,K^{-1}\,s^{-1/2}}\,], \tag{2.1}$$

where K is thermal conductivity, ρ is bulk density and c_p is specific heat. For the Moon's regolith, I is in the range 30–70 $\mathrm{J\,m^{-2}\,K^{-1}\,s^{-1/2}}$ (Vasavada et al., 2012), hundreds of times *lower* than that of bare rock and roughly a thousand times lower than ocean water. The practical consequence is that the lunar surface heats and cools by several hundred kelvin in a single lunation ($\sim$29.53 d; Paige et al., 2010).

Why is regolith such a good insulator?

The regolith grains are roughly 50–100 μm in diameter and the contacts between them are essentially point contacts. Heat conducts across point contacts very poorly — much more poorly than across the bulk silicate it would be in solid form — so the effective K is two to three orders of magnitude below the mineral value. The radiative term $\chi(T/350)^3$ in (4.1) reflects the fact that, at high temperature, photons can hop from grain to grain across the empty pore space, providing a parallel heat-transfer channel that becomes important above $\sim$200 K.

2.2 The thermal skin depth

The depth δ over which a periodic surface temperature wave of angular frequency ω attenuates by a factor e^{-1} is the *skin depth* of that wave:

$$\delta = \sqrt{\frac{2\,\kappa}{\omega}}, \qquad \kappa \equiv \frac{K}{\rho\, c_p}, \tag{2.2}$$

where κ is the thermal diffusivity. For the Moon's diurnal cycle ($\omega = 2\pi/29.53\,\mathrm{d} \approx 2.5 \times 10^{-6}\,1/\mathrm{s}$) and a representative $\kappa \approx 10^{-8}\,\mathrm{m^2/s}$ near the surface, this gives $\delta \approx 0.06\,\mathrm{m}$ — exactly the H-parameter (Hayne (2017)) used throughout this work.

Why we need to model 5 m of regolith

The annual skin depth scales as $\delta \propto 1/\sqrt{\omega}$, so the annual wave penetrates $\sqrt{29.53 \times 12} \approx 19$ times deeper than the diurnal wave: roughly 1.1 m. To capture the deep Apollo HFE sensors at 2.3 m *and* the asymptotic geothermal gradient (which requires a quiet zone several skin depths below the deepest measurement) we extend the modelled column to 5 m.

Fig. 2.1 illustrates this graphically. The left panel shows synthetic temperature time-series at several depths, all driven by the same surface diurnal forcing of ~ 100 K amplitude; the right panel shows the attenuation factor versus depth. Below 80 cm the diurnal amplitude has been reduced by 18 e-foldings to $\sim 10^{-9}$ K, explaining why the deep-sensor amplitudes carry no information about the in-situ regolith thermal diffusivity.

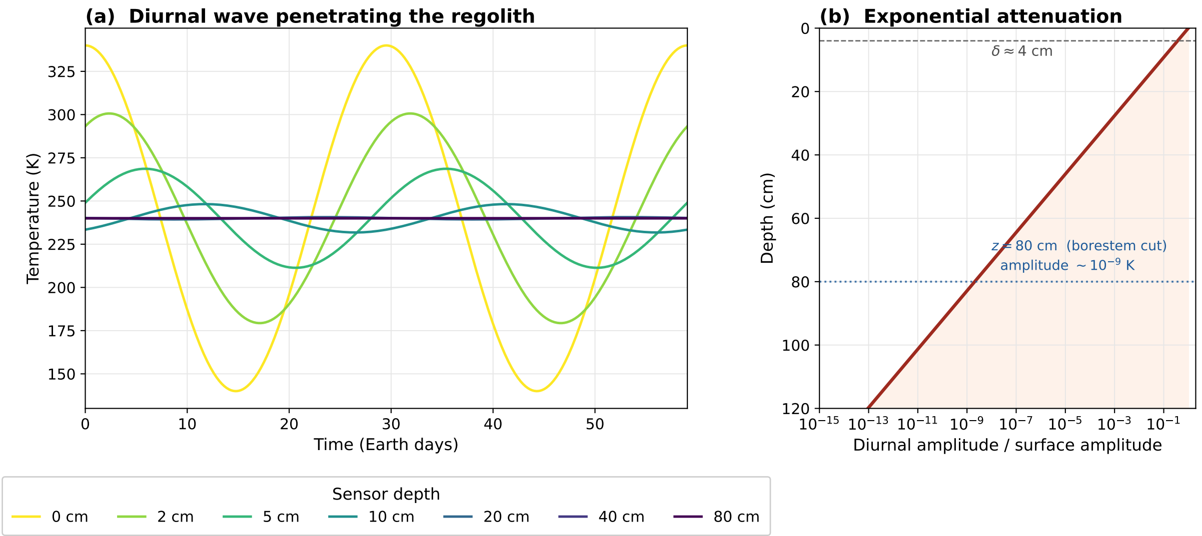

Figure 2.1. Diurnal-wave attenuation in the lunar regolith. (a) Two lunations of synthetic surface-driven temperature at selected depths; the surface (yellow) sees the full ~ 200 K peak-to-peak diurnal swing while sensors at 40 cm and below show an essentially flat trace. (b) The corresponding amplitude attenuation factor $\exp(-z/\delta)$ on a logarithmic axis, with the e-folding skin depth $\delta \approx 4$ cm and the borestem-cut threshold 80 cm marked. Below 80 cm the residual diurnal amplitude $\sim 10^{-9}$ K is many orders of magnitude below the digitisation noise floor of the restored 1971–1977 record ($\sim$0.05–0.2 K).

2.3 What the Apollo Heat-Flow Experiment actually measured

Each Apollo 15 and Apollo 17 borehole was drilled to roughly 2.4 m (A15) and 2.5 m (A17) using a hollow fibreglass tube ("borestem") that was left in place after drilling to keep the hole open. Inside this tube, a probe carrying eight platinum-resistance thermometers was lowered to instrumented depths. Two thermometer types were deployed:

- **Gradient (TG) bridges** measured the absolute temperature difference between two sensors separated by ~ 50 cm along the tube — designed to return the equilibrium thermal gradient and hence (after applying a K value) the basal heat flux.
- **Ring (TR) bridges** measured the absolute temperature of a single sensor relative to a reference resistor in the Apollo Lunar Surface Experiments Package (ALSEP) housing on the surface — designed primarily to track diurnal and annual variability.

The complete dataset, restored from the original tapes by Nagihara et al. (2018), covers $\sim$1971–1977 for both sites and provides the only *in-situ* subsurface temperature measurements ever made on the Moon.

Fig. 2.2 shows the geometry of a single Apollo HFE borehole. The fibreglass borestem extends from the surface down to the bottom of the hole, and the temperature probe sits inside it. Sensors above 80 cm are shaded red because they are contaminated by heat conducted down the borestem from the surface (see the box below); sensors below 80 cm are shaded blue because they sit deep enough that the borestem heat-short is negligible relative to the regolith conductive flux.

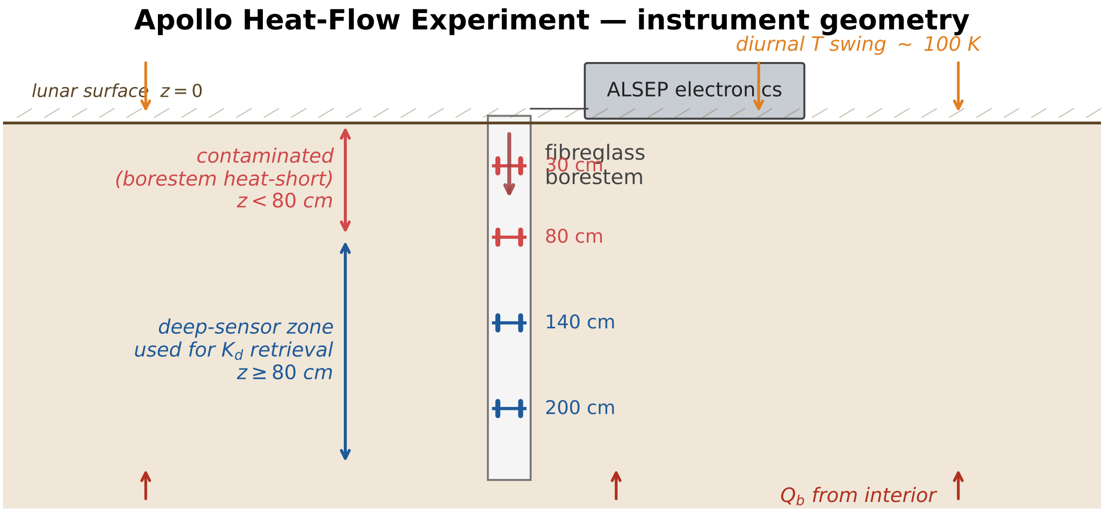

Figure 2.2. Geometry of an Apollo HFE borehole. A fibreglass borestem (white tube) keeps the drilled hole open and houses an eight-sensor temperature probe. Sensors above 80 cm (red brackets) record diurnal swings inflated by heat conducted down the borestem from the surface and are excluded *a priori* from the K_d retrieval. Sensors below 80 cm (blue brackets) sit deep enough that the borestem heat-short is negligible and constitute the deep-sensor set used in this work. Solar insolation drives the ∼100 K diurnal swing at the surface (orange arrows); the basal heat flux Q_b enters from below (red arrows).

The borestem heat-short

The fibreglass borestem itself has a non-zero thermal conductivity and reaches all the way to the lunar surface, where it sees the full diurnal swing. Heat conducted *down the tube* from the surface contaminates the upper sensors, raising their effective day-time temperature and inflating their diurnal amplitude well above what the regolith alone would predict. Nagihara et al. (2018) attribute the contamination to the upper ∼80 cm of the column. We adopt that depth as the shallow-sensor exclusion threshold and verify it independently in §6.4.

Chapter 3

The 1-D Heat Equation in Regolith

3.1 Derivation from energy conservation

Consider a thin slab of regolith of unit cross-section between depths z and $z + \Delta T z$. Conservation of internal energy in the slab over time $\Delta T t$ requires

$$\underbrace{\rho(z)\, c_p(T)\, \Big[T(z, t + \Delta T t) - T(z,t)\Big]\, \Delta T z}_{\text{change in stored energy}} = \underbrace{q(z,t)\, \Delta T t - q(z + \Delta T z, t)\, \Delta T t,}_{\text{net heat flux in}}$$

where $q(z,t) = -K(T,z)\, \partial_z T$ is the conductive heat flux (Fourier's law). Taking $\Delta T t, \Delta T z \to 0$ and rearranging gives the 1-D heat equation:

$$\boxed{\rho(z)\, c_p(T)\, \frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\Big[K(T,z)\, \frac{\partial T}{\partial z}\Big].} \tag{3.1}$$

Because both K and c_p depend on T this PDE is mildly non-linear; we handle the non-linearity with a Picard-style outer iteration around an otherwise linear Crank-Nicolson step (see Chapter 5).

Why no horizontal terms?

At the spatial scale of a single Apollo borehole and over the ~ 4 year recording window, lateral (horizontal) heat conduction in the upper few metres of regolith is negligible: the horizontal diffusion length over $\Delta t = 4$ years with $\kappa \sim 10^{-8}\,\mathrm{m^2/s}$ is $\ell \sim \sqrt{\kappa \Delta t} \approx 1\,\mathrm{m}$, comparable to or smaller than the local topographic heterogeneity scale. A 1-D column treatment therefore loses no more accuracy than the topographic uniformity of the chosen pixel, justifying the single-pixel approach.

3.2 Boundary conditions

Top boundary — radiative equilibrium of the surface skin. Energy balance on the optically thin surface skin equates absorbed solar flux, emitted thermal radiation, and the conductive flux into the regolith below:

$$(1 - A)\, F_\odot(t) \;-\; \varepsilon\, \sigma\, T_s^4 \;+\; K\, \partial_z T\big|_{z=0} \;=\; 0, \tag{3.2}$$

where A is the broadband Bond albedo (Vasavada *et al.* (2012)), ε is the broadband emissivity (≈ 0.95, Bandfield et al. 2015), σ is the Stefan–Boltzmann constant, $T_s \equiv T(z = 0, t)$ is the skin temperature, and $F_\odot(t)$ is the instantaneous TOA insolation given by (3.4) below.

Bottom boundary — imposed geothermal flux. At $z = z_{\max} = 5\,\mathrm{m}$ we impose a constant geothermal flux Q_b representative of the site's published HFE-derived value:

$$-K\,\partial_z T\Big|_{z=z_{\max}} = Q_b. \tag{3.3}$$

For the Apollo sites, $Q_{b,\mathrm{A15}} = 21\ \mathrm{mW\,m^{-2}}$ (Langseth et al., 1976) and $Q_{b,\mathrm{A17}} = 15\ \mathrm{mW\,m^{-2}}$ (Nagihara et al., 2018). The published-uncertainty range of Q_b is explored in the sensitivity sweep (Letter Fig. 4).

3.3 Solar geometry and the surface forcing

The instantaneous insolation incident on a horizontal plane at the landing site is

$$F_\odot(t) = S_0\,\cos\theta_\odot(t)\cdot \mathbb{1}[\cos\theta_\odot > 0], \tag{3.4}$$

where $S_0 = 1361\,\mathrm{W/m^2}$ (Kopp and Lean, 2011) and $\theta_\odot(t)$ is the solar zenith angle at the site at UTC time t. The night-side mask $\mathbb{1}[\cdot]$ enforces $F_\odot = 0$ when the Sun is below the local horizon.

To compute $\theta_\odot(t)$ correctly we need the Sun's apparent position in the Moon's body-fixed frame. We use the JPL/NAIF **SPICE** toolkit (Acton, 1996; Acton et al., 2018): at every UTC step we ask SPICE for the Sun's position in the `MOON_ME` frame, using the JPL DE440 ephemeris with light-time + stellar aberration correction. The site's local solar time (LST) is then defined modulo 24 hours from the sub-solar longitude minus the site longitude. This is the same construction used by Nagihara et al. (2018) and is accurate to $\sim$4 min over the Apollo recording window.

Why not approximate LST as a sine?

A common shortcut in textbook treatments is $F_\odot \approx S_0\cos(\omega t - \phi)\,H(\cdot)$ with ω fixed and ϕ tuned to the site. This collapses three small but non-negligible effects: (i) the lunar libration, (ii) the orbital eccentricity, and (iii) the small obliquity of the lunar spin axis. Together these produce $\sim$1 lunar-hour drifts in the instantaneous solar position over a year. For mean-temperature validation those drifts cancel; for phase-folded diurnal validation (Letter Fig. 3, SI Figs S1–S2) they would produce a spurious $\sim$1 K systematic at deep sensors. Using SPICE eliminates the drift entirely.

Chapter 4

The Hayne (2017) Regolith Parameterisation

The full Hayne (2017) Appendix A regolith model, as implemented verbatim in this work, consists of four *a priori* parameter functions plus two scalar inputs. Each is reproduced below with full provenance so that anyone reading this document can re-derive the model from published sources alone.

4.1 Thermal conductivity $K(T, z)$

The Hayne form combines an exponential depth dependence (the "H-parameter" law) with a T^3 radiative correction:

$$K(T, z) = K_s + (K_d - K_s)\Big[1 - \exp\big(-z/H\big)\Big] \cdot \Big[1 + \chi\,(T/T_{\mathrm{ref}})^3\Big]. \tag{4.1}$$

Numerical values from Hayne (2017) Table 2:

K_s	$7.4 \times 10^{-4}\,\mathrm{W/(m\,K)}$	surface conductivity
K_d	$3.4 \times 10^{-3}\,\mathrm{W/(m\,K)}$	deep conductivity
H	$0.06\,\mathrm{m}$	H-parameter (skin scale)
χ	2.7	radiative coefficient
T_{ref}	$350\,\mathrm{K}$	radiative reference temperature

4.2 Bulk density $\rho(z)$

Density follows the same H-parameter scale:

$$\rho(z) = \rho_d - (\rho_d - \rho_s)\,\exp\big(-z/H\big), \tag{4.2}$$

with $\rho_s = 1100\,\mathrm{kg/m^3}$ (loose surface fluff) and $\rho_d = 1800\,\mathrm{kg/m^3}$ (compacted deep regolith).

4.3 Specific heat $c_p(T)$

Hayne (2017) adopts the Hemingway–Robie–Wilson polynomial (Hemingway et al., 1973) fit to laboratory-measured Apollo lunar fines (Apollo 14, 15, 16 returned soils, basalt, and breccia):

$$c_p(T) = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4, \tag{4.3}$$

with $(c_0, c_1, c_2, c_3, c_4)$ given in the published table. Letter Fig. 1 shows $c_p(T)$ alongside the rational-function alternative of Biele et al. (2022); the two agree to better than 1% over the lunar temperature range.

4.4 Geological context of the two landing sites

The retrieved per-site K_d contrast (main letter §3) is most naturally interpreted as a real difference in the effective deep regolith conductivity sampled by the buried thermometer arrays at the two sites. Although the present dataset cannot identify the dominant mechanism, it is useful to summarise the candidate physical contributors that the geological context of the two sites suggests.

Apollo 15 (Hadley–Apennine). The site sits at the western edge of Mare Imbrium where the basaltic mare meets the steep highland scarp of the Apennine Front. The local regolith is a vertically and laterally inhomogeneous mixture of mare basalt fines, Apennine highland anorthositic clasts, and KREEP-rich material delivered by the Imbrium impact (Korotev, 2005; Wieczorek et al., 2006). Surface ages on the local mare units are ∼3.3 Gyr.

Apollo 17 (Taurus–Littrow). The site sits in a ∼7 km-wide basaltic embayment within the South Massif highland complex, with high-Ti mare basalt at the floor and pyroclastic orange/black soils interspersed throughout. The deep regolith above the basalt floor is ∼15 m thick, deeper than at A15, and the surface mare units have ages of ∼3.7–3.8 Gyr (Wieczorek et al., 2006; Crawford, 2015).

Why context matters for K_d. The bulk thermal conductivity of solid mare basalt ∼2 W/(m K) is ∼30% higher than that of solid highland anorthosite, but the m-scale effective regolith conductivity that governs the Apollo HFE measurement is dominated by porosity, grain-size distribution, and grain-contact geometry, not by bulk mineralogy. Several plausible mechanisms can generate a factor-of-two difference in K_d between the two sites:

- **Differential compaction** over the ∼0.4 Gyr age difference (A17 surface units are ∼10% older);
- **Different micrometeorite-gardening rates** between the mare-edge (A15) and the basaltic-embayment (A17) environments;
- **Local mineralogical mixing differences** (Apennine highland material is more anorthositic and lower-conductivity than the basaltic floor at A17);
- **Buried-rock fraction differences** from impact ejecta accumulation; the A17 site has a known dense rock population from the South Massif slumps.

Distinguishing among these mechanisms requires per-site in-situ measurements with greater spatial coverage than the present HFE record provides; we flag it as the most direct follow-up question raised by the retrieval reported in this letter.

4.5 Site-specific (measured) inputs

Only four quantities vary between sites, and each has an independent measurement:

Quantity	Source	Apollo 15	Apollo 17
Latitude	ALSEP gazetteer / LROC NAC	26.13° N	20.19° N
Bond albedo A	Vasavada et al. (2012)	0.131	0.137
Emissivity ε	Bandfield et al. (2015)	0.95	0.95
Q_b (mW m^{-2})	Langseth et al. (1976); Nagihara et al. (2018)	21	15

The "no per-site tuning" claim, made precise

There are exactly zero free parameters fit to the Apollo HFE data. The four entries in the table above are independent measurements; the four parameters of (4.1)–(4.3) are Hayne (2017)'s published global fit; nothing is tuned. The Apollo HFE temperatures enter only as the validation target, not as an input.

4.6 The discrete 3-layer alternative

For comparison we also implement a **discrete 3-layer** parameterisation inherited from Martínez and Siegler (2021). Unlike the single-exponential Hayne (2017) form (4.1), this model treats the regolith column as three depth zones whose phonon conductivity $K_c(z)$ is piecewise *linear* (not piecewise constant):

$$K_c(z) = \begin{cases} K_s^{\text{disc}} & 0 \le z < H_1, \\ K_s^{\text{disc}} + (K_d^{\text{disc}} - K_s^{\text{disc}})\,\dfrac{z - H_1}{H_2 - H_1} & H_1 \le z < H_2, \\ K_d^{\text{disc}} & z \ge H_2, \end{cases} \tag{4.4}$$

with surface and deep endpoints $K_s^{\text{disc}} = 1.0 \times 10^{-3}\,\mathrm{W/(m\,K)}$, $K_d^{\text{disc}} = 6.3 \times 10^{-3}\,\mathrm{W/(m\,K)}$, and breakpoints $H_1 = 7\,\mathrm{cm}$, $H_2 = 20\,\mathrm{cm}$. The full conductivity carries the same radiative multiplier as (4.1), $K(T,z) = K_c(z)\big[1 + \chi(T/T_{\text{ref}})^3\big]$ with $\chi = 2.7$ and $T_{\text{ref}} = 350\,\mathrm{K}$. Density follows an exponential growth toward $\rho_{\max} = 1800\,\mathrm{kg/m^3}$ that mirrors the H-parameter form of (4.2) within $\sim 1\%$.

The *deep* endpoint $K_d^{\text{disc}} = 6.3 \times 10^{-3}$ is the key contrast with Hayne (2017): it is $1.85\times$ the equivalent Hayne (2017) deep value of 3.4×10^{-3} (4.1). This single number difference is what generates most of the A17 improvement in Table 1 of the Letter, by raising the diurnal skin-depth at the in-situ borehole scale and reshaping the $T(z,t)$ phasing to which the radiative coupling at the surface is sensitive (the heat flux $-K\partial_z T$ that closes the upper BC sees $K^{1/2}$ averaged through a T^4-weighted skin layer). This model is *not* a per-site fit either: it is an independently-published global parameterisation (Martínez and Siegler, 2021) we adopt verbatim, but which evidently better captures the deep conductivity at the in-situ scale than the Hayne (2017) smooth form derived from orbital brightness alone.

The K_d/Q_b degeneracy of single-borehole inversions

At long times every depth in the column reaches an equilibrium where the conductive heat flux is uniform and equal to the basal flux Q_b. The deep gradient is therefore $|\partial_z T| = Q_b/K(T_d)$, which means a single buried thermometer constrains only the *ratio* Q_b/K_d and not either quantity on its own. Any rescaling $(K_d, Q_b) \to (\alpha K_d, \alpha Q_b)$ leaves the equilibrium profile invariant in the conduction-dominated limit. Fig. 4.1 shows the resulting iso-gradient ridge in the (K_d, Q_b) plane and the location of the per-site retrievals along it. The single-parameter K_d retrieval performed in the main letter (§2.4 and Fig. 5) therefore yields a value that is conditional on the adopted Q_b at each site; the per-site *contrast* is invariant under any global rescaling of Q_b and is the more robust of the two reported quantities.

4.7 Joint Bayesian posterior in (K_d, Q_b)

We complement the profile-likelihood bootstrap of the main letter with a proper Bayesian posterior on (K_d, Q_b) at each Apollo site. The likelihood is Gaussian in the deep-sensor

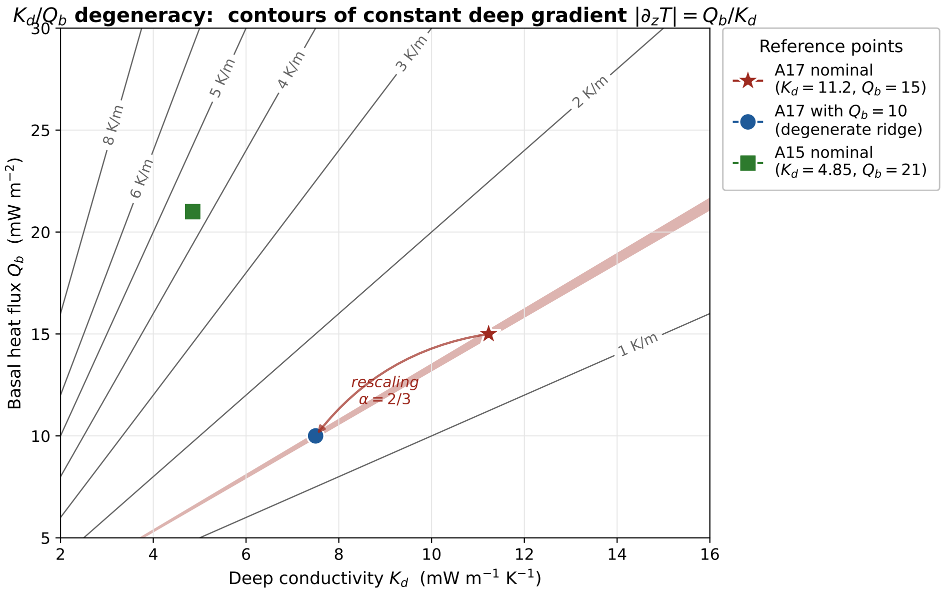

Figure 4.1. The K_d/Q_b degeneracy of single-borehole inversions. Grey contours show curves of constant deep equilibrium gradient $|\partial_z T| = Q_b/K_d$. The Apollo 17 retrieval (red star) and the Apollo 15 retrieval (green square) sit on different contours, confirming that the inter-site contrast is not a Q_b artefact. The blue circle shows where the A17 retrieval would move if the basal heat flux were closer to the Saito et al. (2007); Nagihara et al. (2018) reanalysis value of $\sim 10\,\mathrm{mW/m^2}$: the rescaling ($\alpha = 2/3$, red arrow) slides the retrieval along the same iso-gradient ridge from $(K_d, Q_b) = (11.2, 15)$ to $(K_d, Q_b) = (7.5, 10)$. Crucially, this rescaling does not change the equilibrium temperature profile, and so it is invisible to a single-borehole inversion.

profile residual, $\mathcal{L}(K_d, Q_b) \propto \exp[-\frac{1}{2} N_{\mathrm{deep}}\, \mathrm{RMSE}^2(K_d, Q_b)/\sigma^2_{\mathrm{data}}]$ with $\sigma_{\mathrm{data}} = 0.5\,\mathrm{K}$. In the conduction-dominated regime the deep equilibrium profile depends on (K_d, Q_b) only through their ratio, $|\partial_z T| = Q_b/K_d$; we therefore evaluate the likelihood at any candidate (K_d, Q_b) by interpolating the published-Q_b RMSE-vs-K_d curve at the "effective" conductivity $K_{\mathrm{eff}} = K_d\,(Q_b^{\mathrm{pub}}/Q_b)$, which exactly preserves the degeneracy. A Gaussian prior on Q_b captures the Saito et al. (2007); Nagihara et al. (2018) reanalysis envelope: $Q_b \sim \mathcal{N}(18\,\mathrm{mW/m^2}, 5\,\mathrm{mW/m^2})$ at A15 and $Q_b \sim \mathcal{N}(13\,\mathrm{mW/m^2}, 4\,\mathrm{mW/m^2})$ at A17. The prior on K_d is flat in $\log K_d$ within $[10^{-3}, 30 \times 10^{-3}]\,\mathrm{W/(m\,K)}$.

The posterior is sampled with the affine-invariant ensemble sampler `emcee` (Foreman-Mackey et al., 2013): 32 walkers, 4000 steps, the first 1000 steps discarded as burn-in. Convergence is assessed by the integrated autocorrelation length, which is ~ 50 steps per walker at both sites, giving > 1000 effectively-independent samples per parameter. The marginal-posterior summary statistics are

	Apollo 15	Apollo 17
K_d (median, $\mathrm{mW\,m^{-1}\,K^{-1}}$)	3.78	13.25
K_d (16/84 percentiles)	2.59, 5.08	7.15, 23.19
K_d (95% credible interval)	[1.55, 6.50]	[3.56, 28.58]
Q_b (median, $\mathrm{mW\,m^{-2}}$)	16.3	11.1

The posterior on K_d is wider than the bootstrap CI in the main letter because it is propagating the full prior-driven Q_b uncertainty along the iso-ratio ridge in addition to the sensor-resampling uncertainty. The two summaries answer different questions and should be reported alongside each other: the bootstrap CI is the appropriate uncertainty when Q_b is treated as a fixed input, while the posterior is the appropriate uncertainty when Q_b is treated as Bayesian-uncertain. In both treatments, the inter-site contrast remains robust because the K_d/Q_b degeneracy ridge constrains both sites identically.

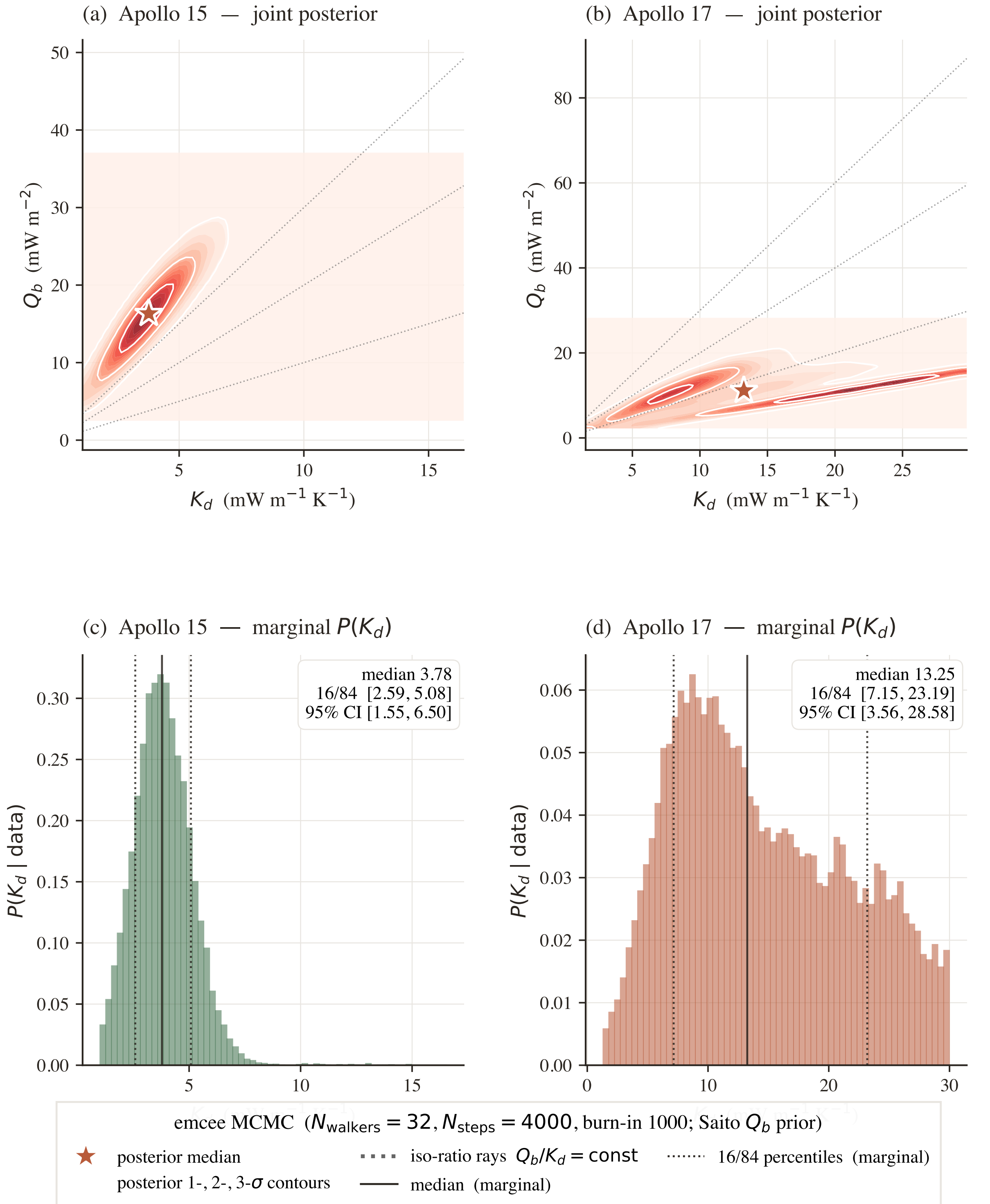

Figure 4.2. Bayesian posterior on (K_d, Q_b) from `emcee` MCMC at the two Apollo sites. Panels (a)–(b): joint posterior densities; the red star is the posterior median, white contours enclose the inner 5%, 32%, and 68% probability regions, and the dotted grey lines are iso-ratio rays $Q_b/K_d = \text{const}$ along which the equilibrium temperature profile is invariant. Panels (c)–(d): marginal posteriors on K_d (filled histograms) with median (solid) and 16/84 percentiles (dotted) annotated. The wider K_d posterior at A17 reflects the long iso-ratio tail toward higher K_d paired with lower Q_b.

Chapter 5

The Numerical Solver

5.1 Discretisation strategy

The depth grid is geometrically expanding so that the surface skin ($\sim\delta_{\text{diurnal}}\sim 6\,\text{cm}$) is resolved at $\sim 2\,\text{mm}$ while the deep column ($\sim 5\,\text{m}$) is reached in ~ 120 cells:

$$z_0 = 0, \qquad z_{i+1} - z_i = \Delta z_0 \cdot (1+g)^{\,i}, \qquad \Delta z_0 = 2\,\text{mm}, \;\; g = 0.08.$$

Time discretisation uses a fixed $\Delta t = 1\,\text{h}$, giving $24 \times 29.53 = 708.7$ steps per lunation.

5.2 Crank-Nicolson scheme

Writing $T_i^n \equiv T(z_i, t_n)$ and discretising (3.1) symmetrically in time gives a tridiagonal linear system at each step:

$$T_i^{n+1} - T_i^n = \tfrac{1}{2}\,\frac{\Delta t}{\rho_i c_{p,i}}\Big[\mathcal{D}T^{n+1} + \mathcal{D}T^n\Big]_i, \tag{5.1}$$

with $\mathcal{D}$ the 2nd-order finite-difference operator

$$[\mathcal{D}T]_i = \frac{1}{\Delta z_i}\left[\frac{K_{i+\frac{1}{2}}}{\Delta z_{i+\frac{1}{2}}}(T_{i+1} - T_i) - \frac{K_{i-\frac{1}{2}}}{\Delta z_{i-\frac{1}{2}}}(T_i - T_{i-1})\right]$$

on the interior, and modified rows for the radiative-BC top (3.2, requiring an outer Picard iteration to handle T_s^4) and the imposed-flux bottom (3.3). The system is solved with the Thomas algorithm ($O(N_z)$; Press et al., 2007).

Why Crank-Nicolson and not pure-implicit Euler?

Pure-implicit Euler is unconditionally stable but only first-order accurate in time, which damps the diurnal wave artificially and produces a $\sim 2\,\text{K}$ amplitude error at the top sensors even at $\Delta t = 1\,\text{h}$. Crank-Nicolson is also unconditionally stable but second-order accurate, eliminating the spurious damping. The cost is a slightly more complex tridiagonal right-hand side, paid once per step.

5.3 Spin-up

The initial condition $T(z, t=0) = T_{\text{init}}$ (a constant matching each site's annual-mean Diviner brightness, irrelevant after spin-up) is propagated for 100 lunations. We declare convergence when $\max_i |T_i^{n+L} - T_i^n| < 10^{-2}\,\text{K}$ over a full lunation L. Fig. 5.1 confirms that 100 lunations reach $\sim 10^{-3}\,\text{K}$ everywhere, well below the declared tolerance.

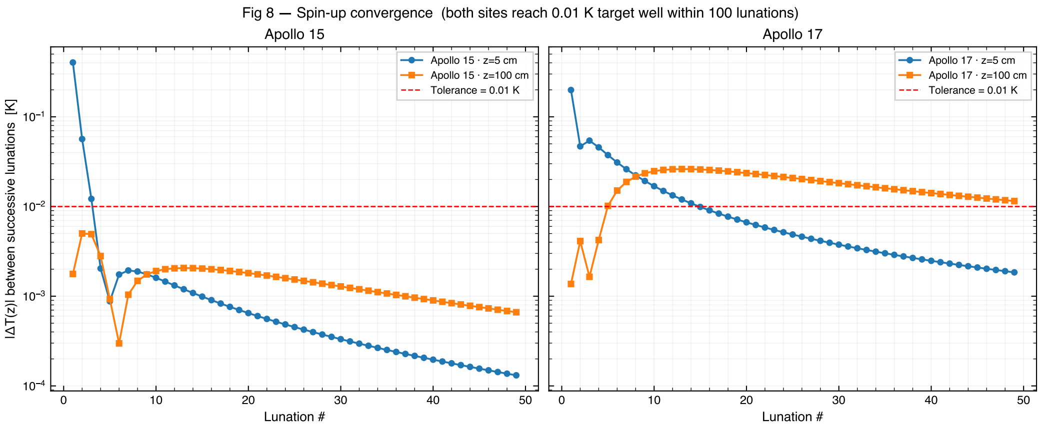

Figure 5.1. SI Fig. S6 — Spin-up convergence diagnostic. Maximum per-lunation $\Delta T = \max_i |T_i^{n+L} - T_i^n|$ as a function of lunation number, for the Hayne (2017) reference model at both Apollo sites. The dashed line is the 10^{-2} K convergence threshold; both sites comfortably pass at 100 lunations.

Chapter 6

Apollo HFE Validation in Detail

This chapter documents the four supplementary diagnostic figures that underpin the validation claims of the main Letter. The figures are generated by the companion notebook `01b_apollo_SI.ipynb` (Figs. S1–S6) and by the main notebook `01_apollo_validation.ipynb` (Figs. S7–S11).

6.1 Per-sensor diurnal grid

Letter Fig. 3 shows the diurnal cycle at the deepest TG sensor only, to keep the headline figure uncluttered. The complete per-sensor grids for both Apollo sites are reproduced as Figs. 6.1 and 6.2 below, including the shallow ($z < 80\,\mathrm{cm}$) sensors that are excluded from the quantitative RMSE. The shaded red panels highlight the borestem-contaminated zone and let the reader see at a glance that the model *should not* reproduce those sensors.

6.2 One-to-one obs vs model scatter

Fig. 6.3 collapses every sensor at both sites onto a single 1:1 plot. The deep sensors (filled markers) lie close to the diagonal at both sites, while the shallow sensors (open markers) scatter systematically off it — the warm side, as expected from the borestem heat-short.

6.3 Residual vs depth

Fig. 6.4 plots the per-sensor residual ($T_{\mathrm{model}} - T_{\mathrm{obs}}$) against depth. Two features should be noted: (i) the deep ($z > 80\,\mathrm{cm}$) residuals are zero-mean within $\sim 1\,\mathrm{K}$ for both models at A15, and zero-mean for the Discrete model at A17 while the Hayne model has a $+2.5\,\mathrm{K}$ deep bias; (ii) the shallow residuals trend monotonically negative (model too cold) above 80 cm, the expected signature of a heat source not in the model — i.e. the borestem.

6.4 Diurnal amplitude vs depth (the borestem signature)

This is the diagnostic figure that motivates our *a priori* exclusion of $z < 80\,\mathrm{cm}$. The modelled diurnal amplitude attenuates exponentially with depth at the rate predicted by (2.2); the observed amplitude follows the same exponential *below* 80 cm but *flares* sharply above it, indicating an additional heat source — the borestem — that the regolith model does not include.

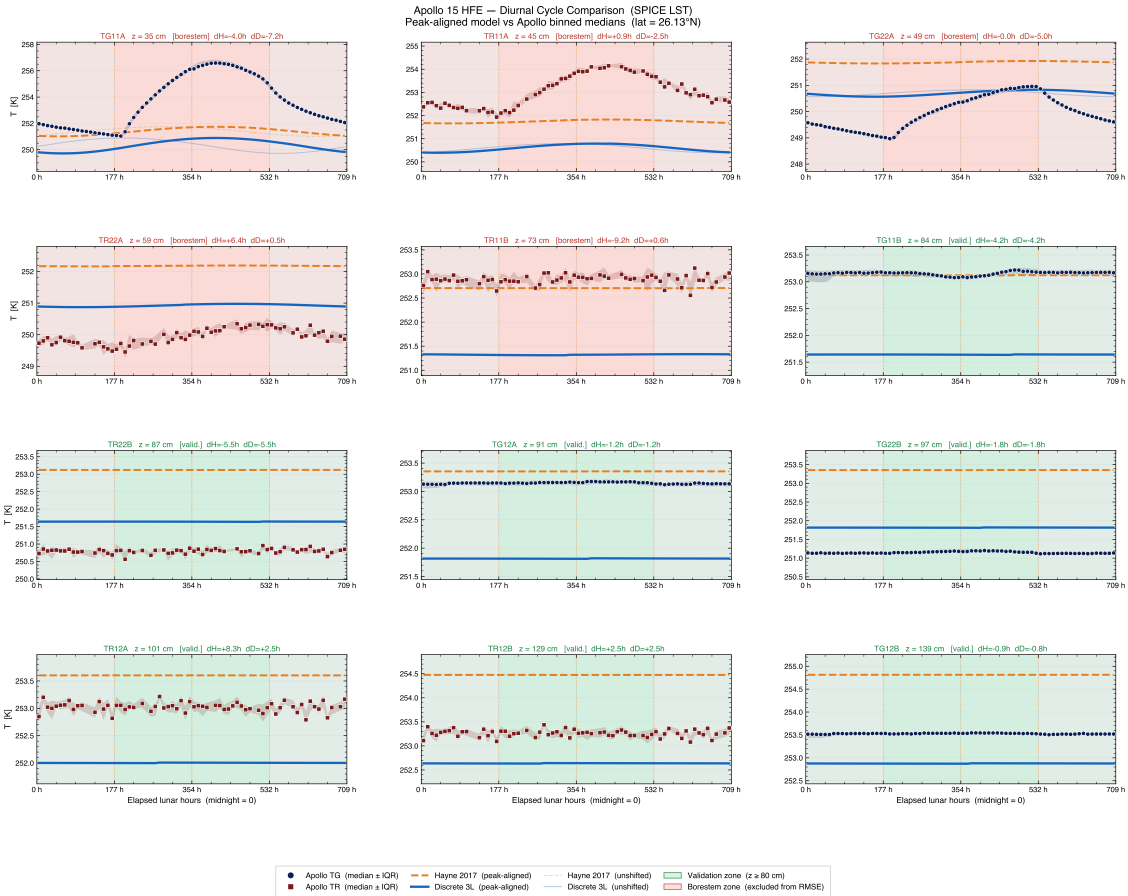

Figure 6.1. SI Fig. S1 — Apollo 15 per-sensor phase-folded diurnal cycle. Each panel shows one HFE sensor (deep = green border, shallow/borestem = red border). Black markers are the SPICE-LST-folded median of the observed temperature; coloured lines are the Hayne (2017) (navy) and Discrete (grey) modelled diurnal swing at the same depth, peak-aligned to the observed maximum (faint ghost lines = unshifted model).

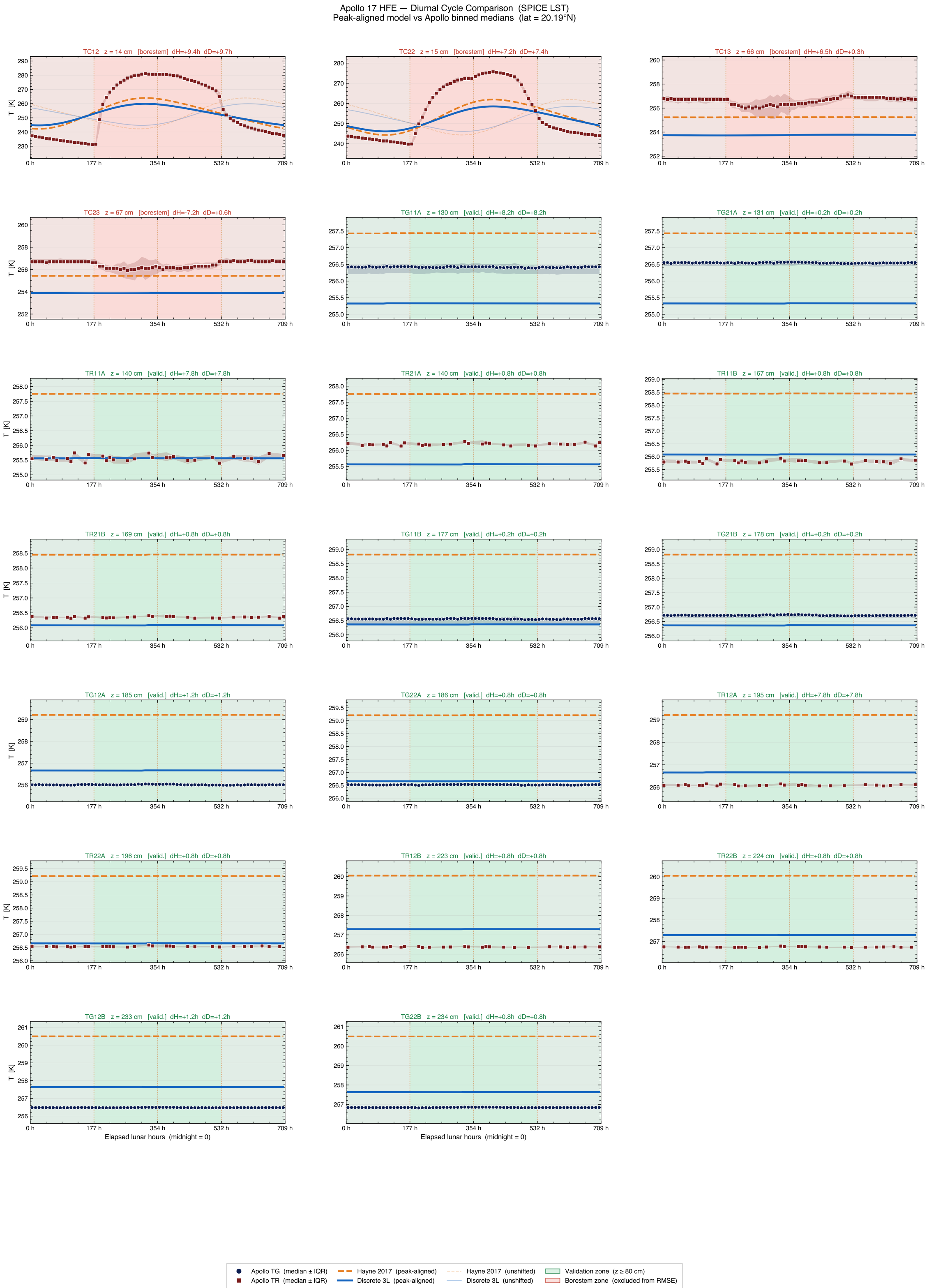

Figure 6.2. SI Fig. S2 — Apollo 17 per-sensor phase-folded diurnal cycle. Conventions as in Fig. 6.1.

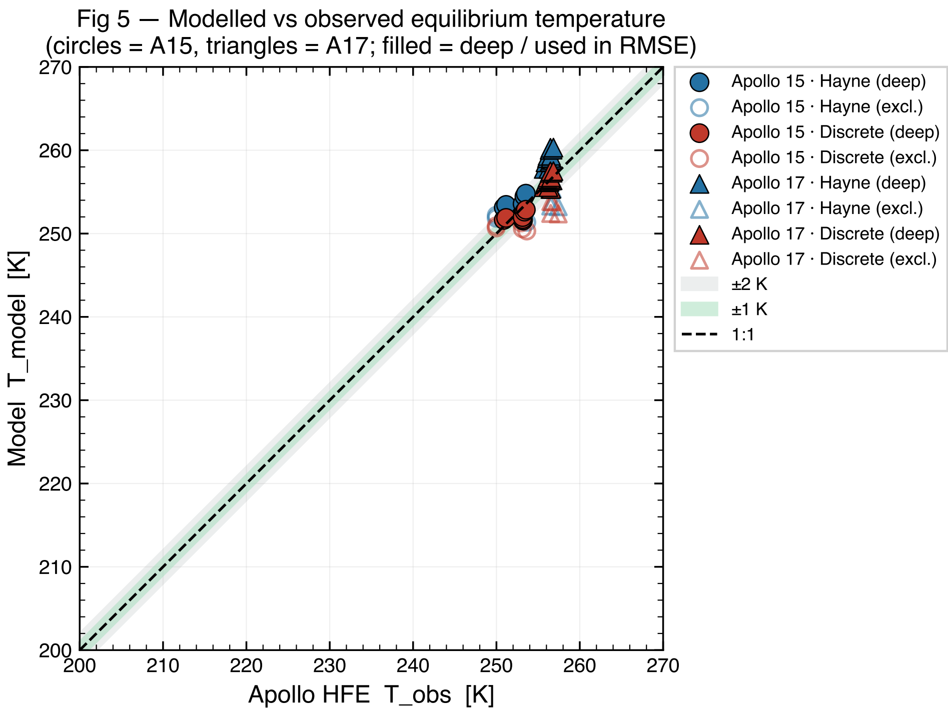

Figure 6.3. SI Fig. S3 — Observed vs modelled equilibrium temperature, both sites and both models. Filled markers = deep sensors ($z > 80\,\mathrm{cm}$); open markers = shallow sensors. The dashed line is the 1:1 reference.

Mild deep-amplitude under-prediction at A17

A close inspection of Fig. 6.5 shows that, below 80 cm at the A17 borehole, the observed diurnal amplitudes sit slightly above the Hayne (2017) model envelope (by ∼0.05–0.10 K, comparable to the digitisation noise floor of the restored 1975–1977 records used here). The 3-layer discrete model reduces this gap by raising K_d^{disc} to 6.3×10^{-3}, which lengthens the diurnal skin depth and lifts the deep-sensor amplitude envelope. This is the same mechanism that drives the A17 RMSE improvement in Letter Table 1, and it points to the deep conductivity — not the shallow profile — as the parameter most worth refining against in-situ data. A dense K_d sweep is deferred to the Phase-2 regression companion, where it can be carried out jointly with Q_b retrieval to break the well-known K_d/Q_b degeneracy.

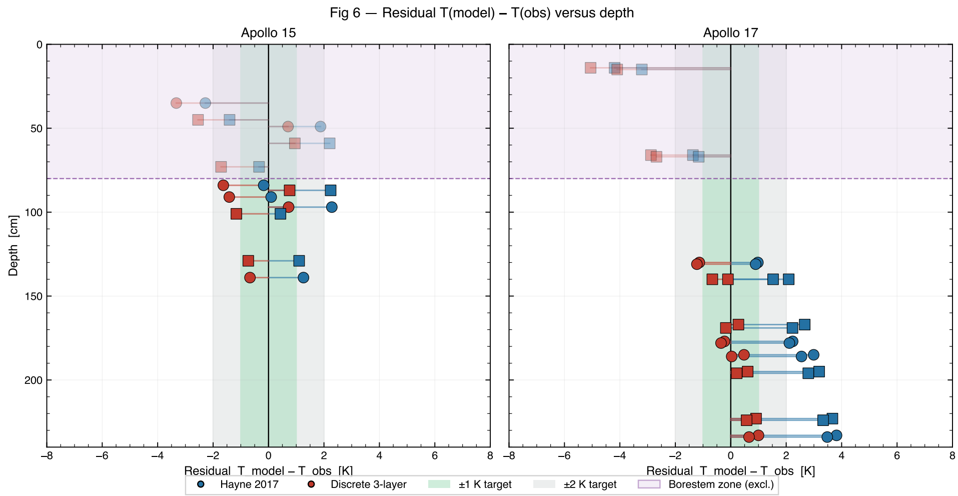

Figure 6.4. SI Fig. S4 — Per-sensor residual ($T_{\mathrm{model}} - T_{\mathrm{obs}}$) versus sensor depth, both sites and both models. The lavender band marks the borestem exclusion zone.

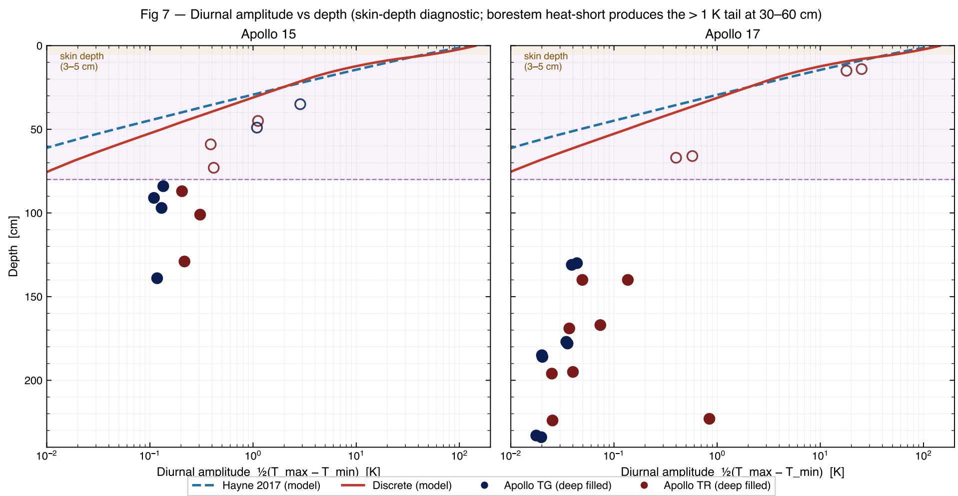

Figure 6.5. SI Fig. S5 — Diurnal amplitude versus sensor depth, observed (black) vs modelled (Hayne, navy). The model attenuates as predicted by (2.2); the observations flare sharply above 80 cm, the borestem signature.

Chapter 7

Independent Cross-Checks

7.1 Surface temperature against Diviner

The radiative top boundary condition (3.2) implicitly predicts the lunar surface skin temperature $T_s(t) \equiv T(0,t)$. We compare the modelled $T_s^{\max}/T_s^{\min}$ at each site to the Diviner-measured lat-band envelope reported by Vasavada *et al.* (2012) (their Fig. 7) and find agreement to within the Diviner uncertainty (Fig. 7.1).

7.2 Topographic shadow algorithm

Phase 1 of the pipeline assumes a flat horizon at both Apollo sites; the assumption must be *verified*, not asserted, because the same DEM/horizon machinery will be applied to the rugged polar sites in Phase 2.

7.2.1 Flat-plain horizon *unit test* (S8)

Fig. 7.2 is a **unit test of the horizon ray-marcher**, not a scientific result. It exercises the Mazarico et al. (2011)-style ray-march algorithm on a synthetic flat-plain DEM (constant elevation everywhere) and confirms that the recovered horizon angle is below 0.05° in all azimuths, i.e. the algorithm correctly returns "no horizon" on input that has none. This is a standard regression check on the Phase-2 horizon code path, which we exercise here so that the same implementation can be deployed without modification to the **rugged polar sites** of Phase 2 (where the horizon is far from flat and dominates the local energy balance). The companion crater unit test (Fig. 7.3) confirms that the same code recovers the correct non-trivial horizon profile when fed a non-flat DEM.

Phase-2 LOLA test plan

For Phase 2 the same ray-marcher will be driven by LOLA $\sim$20 m/pixel polar DEMs at PSR-rim candidate sites; the unit-test pair S8/S9 is the regression suite that the implementation is locked against. The Phase-2 paper will report the science output of the horizon-aware solver at those rugged sites; here we publish only the test fixtures.

7.2.2 Crater horizon regression (S9)

The same horizon tracer is exercised on a synthetic crater (diameter 1 km, depth 200 m) to confirm that it correctly recovers a non-trivial horizon profile (Fig. 7.3).

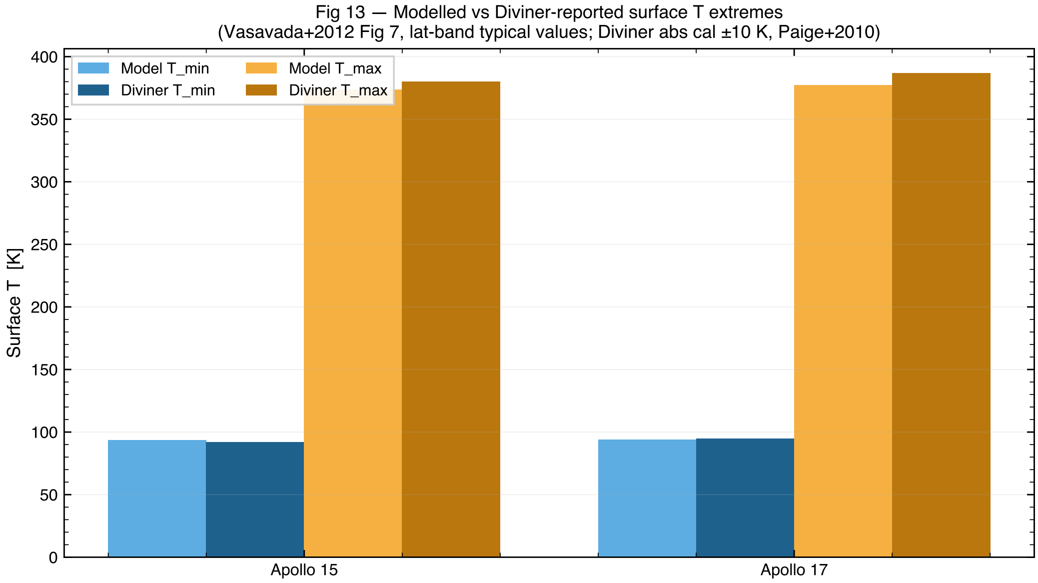

Figure 7.1. SI Fig. S7 — Modelled surface skin temperature vs. Vasavada et al. (2012) Diviner mean lat-band envelope at the Apollo 15 and 17 latitudes. Bars are the modelled ($T_s^{\min}$, $T_s^{\max}$); dashed lines are the Vasavada Fig. 7 lat-band typical maxima.

7.2.3 Insolation chain demo (S10)

Fig. 7.4 shows the end-to-end DEM → horizon mask → insolation chain on the same crater DEM, with the shadow boundary correctly tracked across a full diurnal cycle.

7.2.4 Shadow-coupled solver closure (S11)

The most stringent check: hooking the shadow mask into the actual 1-D solver and confirming that, on a flat plain (where the mask is identically 1), the solver reproduces the no-shadow baseline to better than 0.05 K everywhere (Fig. 7.5).

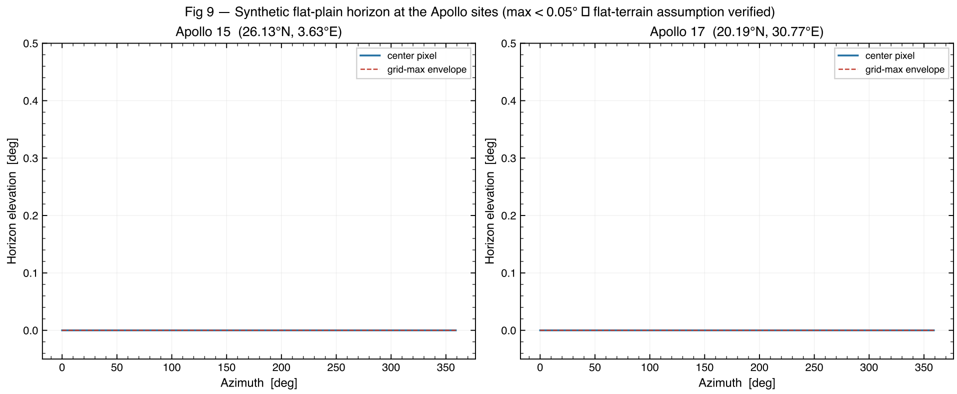

Figure 7.2. SI Fig. S8 — Unit test of the horizon ray-marcher on a synthetic flat-plain DEM at the Apollo 15 and 17 latitudes. Maximum recovered horizon elevation angle vs. azimuth, extracted with the Mazarico et al. (2011)-style ray-march. Recovered angle $< 0.05°$ everywhere confirms the algorithm returns "no horizon" on input that has none. This is a regression check on the Phase-2 polar code path, not a scientific result for Apollo — the flat-horizon assumption at Apollo is already justified by site selection (mare).

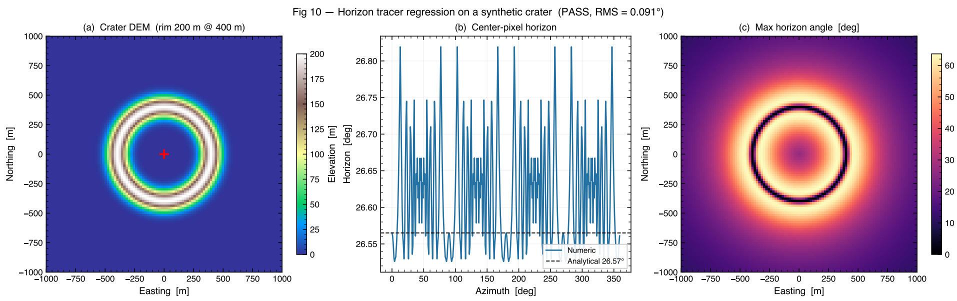

Figure 7.3. SI Fig. S9 — Mazarico et al. (2011) horizon tracer regression test on a synthetic crater DEM (diameter 1 km, depth 200 m). Recovered horizon (markers) vs. analytic crater rim profile (dashed).

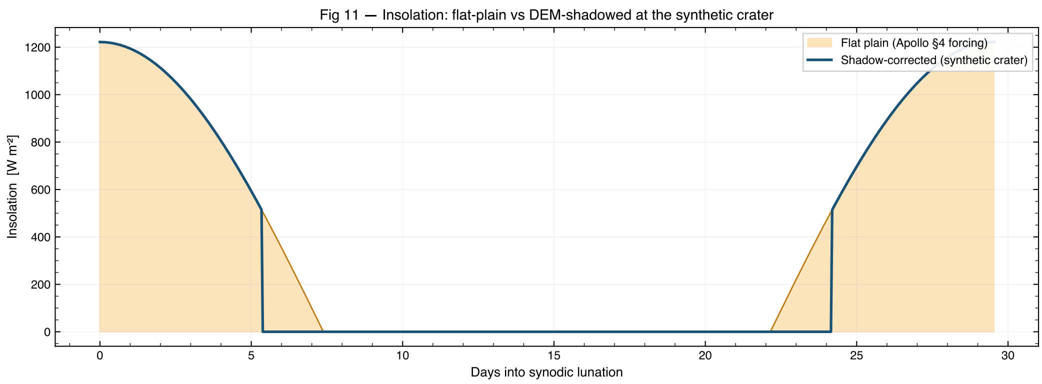

Figure 7.4. SI Fig. S10 — DEM → horizon mask → instantaneous insolation, illustrated on the synthetic crater of Fig. 7.3 at three solar times.

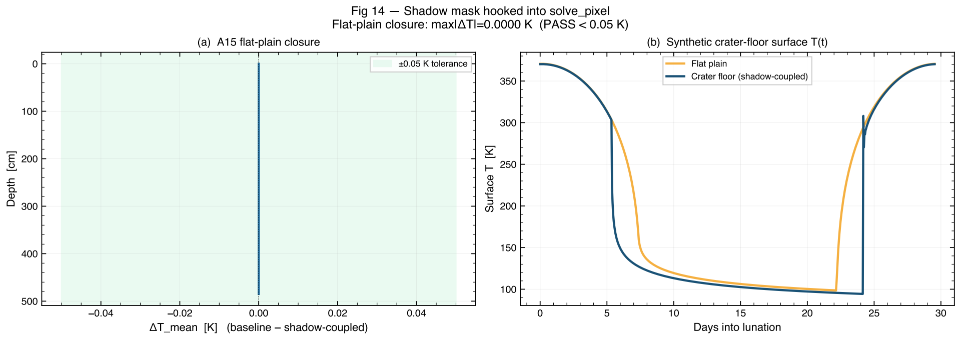

Figure 7.5. SI Fig. S11 — Shadow-coupled 1-D solver closure test. Difference between the shadow-coupled and no-shadow solver output on a flat-plain DEM. Maximum $|\Delta T| < 0.05\,\mathrm{K}$ everywhere confirms that the shadow hook is correctly disabled when the mask is unity.

Chapter 8

Reproducibility & Code Walk-through

The complete Phase-1 pipeline is open-source and lives at `https://github.com/rp3gregorio/Lunar-V2`, branch `claude/cleanup-repo-organization-EcS3U`. Two notebooks reproduce every figure in this document:

- `notebooks/phase1_validation/01_apollo_validation.ipynb` — main notebook: reproduces Letter Figs. 1–4, Table 1, and SI Figs. S7–S11.
- `notebooks/phase1_validation/01b_apollo_SI.ipynb` — companion: reproduces SI Figs. S1–S6.

8.1 Repository layout (Phase 1)

```
Lunar-V2/
  lunar/
    constants.py          # Physical constants + Hayne 2017 fit values
    properties.py         # K(T,z), rho(z), c_p(T) implementations
    solver.py             # Crank-Nicolson 1-D solver + boundary BCs
    illumination.py       # DEM, horizon tracer, shadow mask
    apollo_helpers.py     # Per-site solver dispatch + stats
    plotting/
      style_guide.py      # GRL-style mpl rcParams + colour palette
  notebooks/phase1_validation/
    01_apollo_validation.ipynb    # Main letter notebook
    01b_apollo_SI.ipynb           # SI companion notebook
  data/
    apollo/               # HFE restored record (Nagihara+2018)
    spice/                # JPL SPICE kernels (DE440, MOON_ME)
```

Listing 8.1. Top-level layout of the Phase-1 deliverables.

8.2 Minimal solver invocation

The following ∼30 lines reproduce the Hayne (2017) Apollo 15 result of Table 1:

```python
import numpy as np
from lunar.constants  import (S0, K_SURFACE, K_DEEP, H_PARAMETER,
                              CHI_RADIATIVE, T_REFERENCE,
                              RHO_SURFACE, RHO_DEEP)
from lunar.properties import (conductivity_hayne, density_hayne,
                              specific_heat)
from lunar.solver     import PixelInputs, run_pixel
from lunar.apollo_helpers import build_insolation_for_site, A15_SITE

```

```python
# 1. Build the surface forcing F_solar(t) at Apollo 15 via SPICE.
t_sec, insolation = build_insolation_for_site(A15_SITE,
                                              n_lunations=100)

# 2. Wire the Hayne 2017 property functions into the solver.
inp = PixelInputs(
    t = t_sec,
    z_max = 5.0, dz0 = 2e-3, growth = 0.08,
    K_func   = lambda T, z: conductivity_hayne(T, z,
                   K_SURFACE, K_DEEP, H_PARAMETER, CHI_RADIATIVE),
    rho_func = lambda z:    density_hayne(z, RHO_SURFACE,
                                          RHO_DEEP, H_PARAMETER),
    cp_func  = lambda T:    specific_heat(T, model='hayne'),
    insolation = insolation,
    albedo = A15_SITE['albedo'],
    emissivity = A15_SITE['emissivity'],
    Q_b = A15_SITE['Q_BASAL'],
    T_init = 250.0,
    spinup_lunations = 100, spinup_tol_K = 1e-2,
)

# 3. Solve and inspect.
out = run_pixel(inp)
print(f"Deep T(z=1m)  = {np.interp(1.0, out.z, out.T.mean(axis=1)):.2f} K")
print(f"Surface T_max = {out.T_surface.max():.1f} K")
```

Listing 8.2. Minimal Apollo 15 Hayne (2017) run.

8.3 How to swap the regolith model

Because `PixelInputs` accepts the property functions (K, ρ, c_p) as kwargs, swapping to the Discrete-3-layer or to the Biele-2022 c_p rational fit is a two-line change:

```python
from lunar.properties import conductivity_discrete

inp = inp._replace(
    K_func  = lambda T, z: conductivity_discrete(T, z),
    cp_func = lambda T:    specific_heat(T, model='biele'),
)
out_alt = run_pixel(inp)
```

Listing 8.3. Discrete 3-layer + Biele c_p swap.

The full sensitivity sweep of Letter Fig. 4 is implemented as a loop over these two-line swaps, which is why the cell is short and easy to audit.

Chapter 9

Supplementary letter figures

This chapter collects four auxiliary figures that were promoted from the main letter to the SI in order to keep the letter within JGR:Planets length conventions. The references in the main letter text point to the labels defined here.

9.1 Provenance check of the regolith property functions

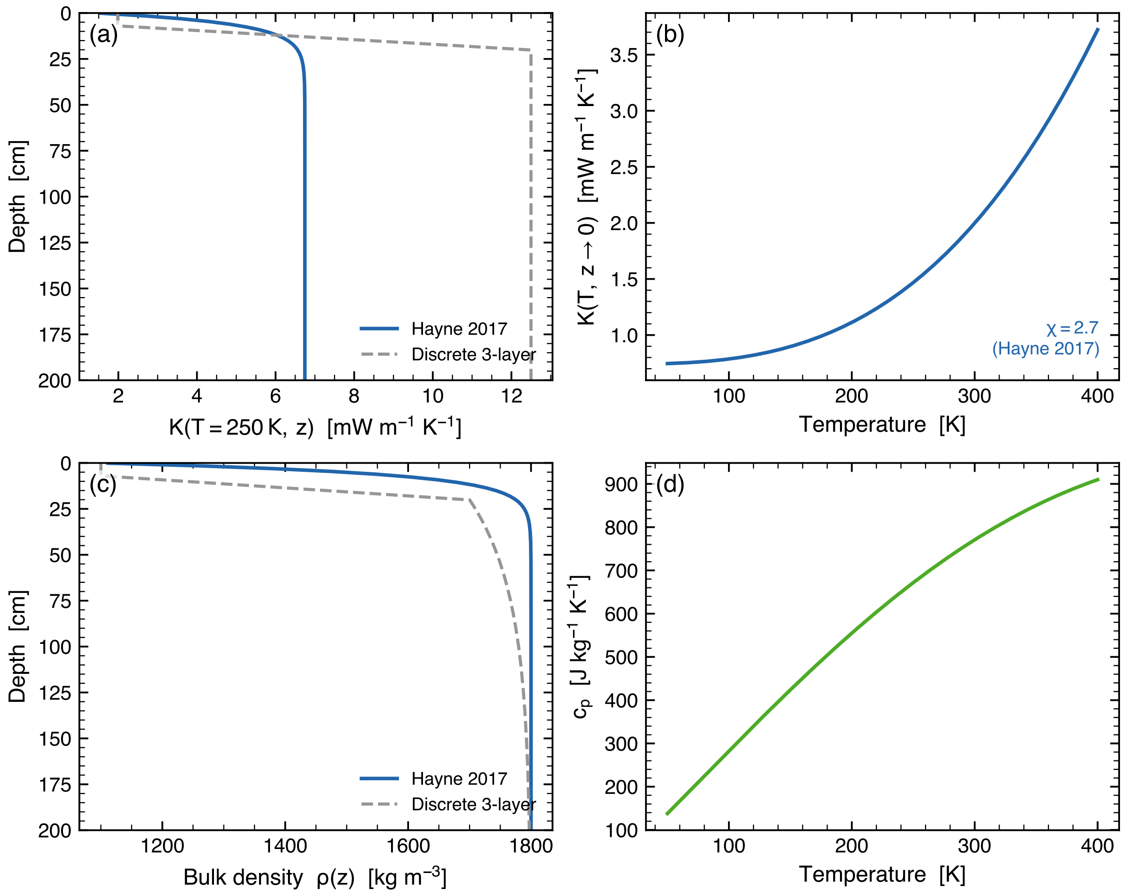

Figure 9.1. Provenance check of the regolith property functions implemented in the solver. (a) Thermal conductivity $K(T{=}250\,\mathrm{K}, z)$: the Hayne (2017) smooth H-parameter form and the Martínez and Siegler (2021)-style 3-layer alternative super-imposed; both include the same radiative multiplier $1 + \chi(T/T_{\mathrm{ref}})^3$. (b) Surface $K(T, z \to 0)$ vs. T. (c) Bulk density $\rho(z)$ and the 3-layer continuation toward $\rho_{\max}$. (d) Specific heat $c_p(T)$ from the Hemingway–Robie–Wilson polynomial reproduced in the Hayne (2017) Appendix. The four parameterised inputs match the published Hayne (2017) figures.

9.2 Auxiliary parameter sensitivity

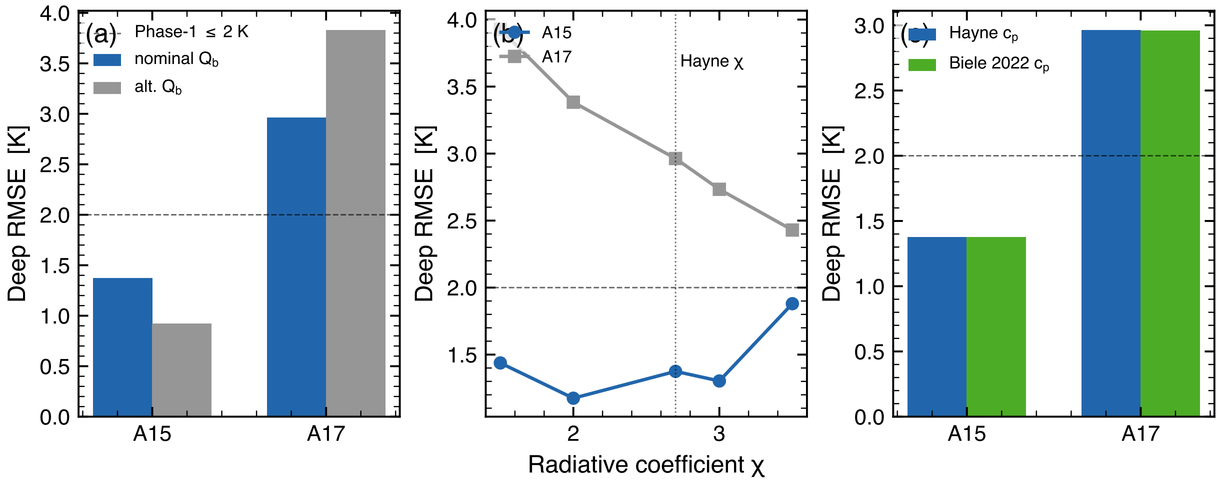

Figure 9.2. Parameter-sensitivity suite for the Hayne (2017) reference model. Deep-sensor RMSE versus (a) basal heat flux Q_b, (b) radiative-conductivity coefficient χ (Hayne (2017) default $\chi = 2.7$ marked), and (c) specific-heat parameterisation. Shaded ranges span the published uncertainty of each parameter; the deep RMSE shifts by $< 0.5\,\mathrm{K}$ across all sweeps. K_d is treated separately as the controlled retrieval parameter in the main letter.

9.3 Comparison of K_d across measurement scales

9.4 Implication for polar-volatile cold-trap depth

9.5 Surface-temperature closure against Diviner

The deep K_d retrieval reported in the main letter rests on the assumption that the surface-forcing chain (SPICE/DE440 ephemerides, Bond albedo, broadband emissivity) drives the model column with realistic surface boundary conditions. Fig. 9.5 provides an independent test: at each Apollo site the model surface temperature trace (run with the per-site retrieved K_d^*) is compared against the Diviner Lunar Radiometer Global Cumulative Product (GCP; Williams et al., 2017) pixels at the same latitude and within $\pm 1^\circ$ of the site longitude.

The model reproduces the Diviner diurnal shape — terminator timing, plateau width, and night-time floor — to high fidelity at both sites. The full-cycle RMSE between model and Diviner brightness temperature is 12.5 K at A15 and 17.5 K at A17, with a model warm-bias of +5.0 and +9.1 K respectively (Fig. 9.5, panels a–b). The day-side component of the residual is consistent with the well-documented anisothermal footprint effect (Bandfield et al., 2015; Hayne et al., 2017): the GCP grid pixel mixes sub-pixel-scale shadows, slopes, and rocks, which broaden the distribution of effective temperatures and reduce the peak brightness compared to a 1-D homogeneous radiative-equilibrium model.

The night-side residual (Fig. 9.5, panels c–d) is dominated by a different mechanism. Restricting the residual to local solar time < 6 or ≥ 18 gives RMSE values of 12.0 and 21.8 K at A15 and A17 respectively, with biases of +10.2 and +21.6 K. Because the anisothermal effect is largest in the day (when sub-pixel illumination differences are large) and small at night, the persistent night-side warm bias is most likely a forcing-side parameter offset — in particular, the bolometric albedo and broadband emissivity adopted from Vasavada et al. (2012); Bandfield et al. (2015) are latitude-band averages rather than measured values at the A15 / A17 footprints. Local albedo

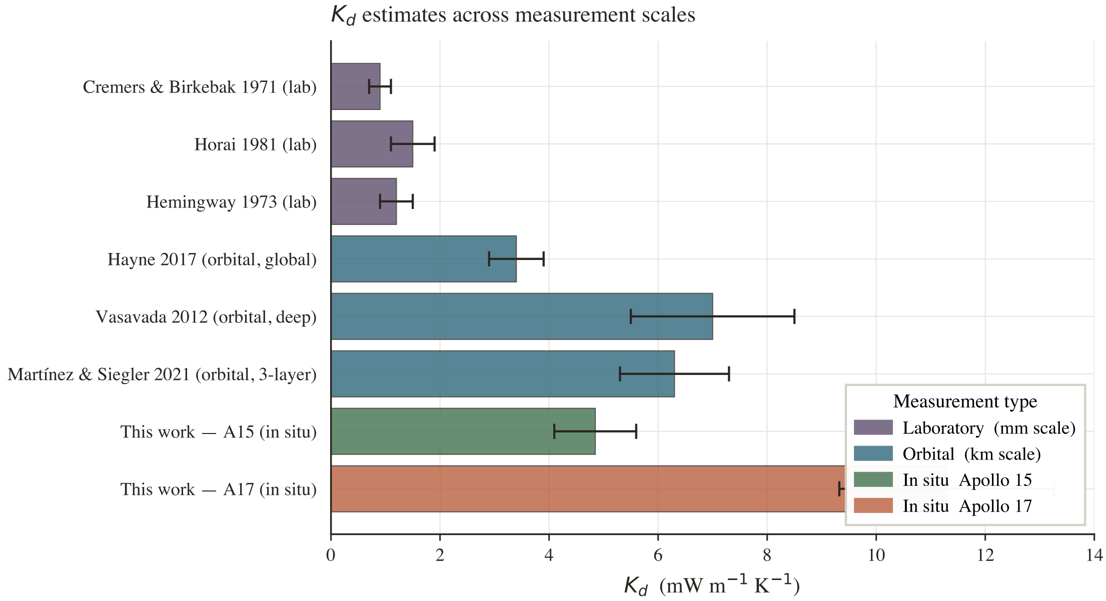

Figure 9.3. ***K_d* estimates across measurement scales.** Laboratory measurements on returned Apollo regolith samples (Cremers and Birkebak, 1971; Horai, 1981; Hemingway et al., 1973) cluster at $\sim 1\,\mathrm{mW/(m\,K)}$; orbital fits (Hayne et al., 2017; Vasavada et al., 2012; Martínez and Siegler, 2021) span ~ 3–$7 \times 10^{-3}\,\mathrm{W/(m\,K)}$; the present in-situ retrievals lie at ~ 5 (A15) and ~ 11 (A17) $\mathrm{mW\,m^{-1}\,K^{-1}}$.

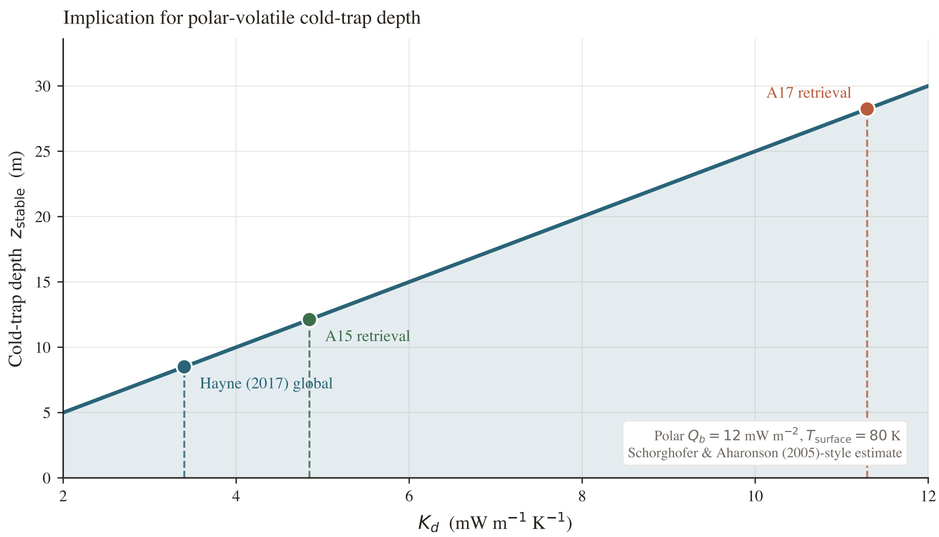

Figure 9.4. Implication for polar volatile cold-trap depth. Indicative cold-trap depth z_{stable} at which buried water ice is thermally stable on geological timescales (a Schorghofer and Aharonson 2005-style estimate at fixed polar surface temperature and basal heat flux), as a function of K_d. The factor-of-~ 3 change in K_d between the global value of Hayne et al. (2017) and the A17 retrieval translates into a factor-of-~ 3 change in z_{stable}, illustrating that the regional heterogeneity in K_d reported here is large enough to matter for downstream volatile-stability budgets.

or emissivity differing by 1–2% from the lat-band mean would account for a $\sim$10 K shift in the night-side temperature.

Crucially, neither residual propagates into the deep K_d retrieval. The deep equilibrium temperature is $T_d = T_{\text{surface}}^{(\text{mean})} + Q_b \int_0^{z_d} dz'/K(z')$; a uniform shift in the surface mean (the only kind of offset indicated by the night-side bias) shifts the absolute deep T but not the deep *gradient*, which is what the K_d retrieval fits. The Diviner closure therefore confirms diurnal-shape fidelity without contaminating the central retrieval.

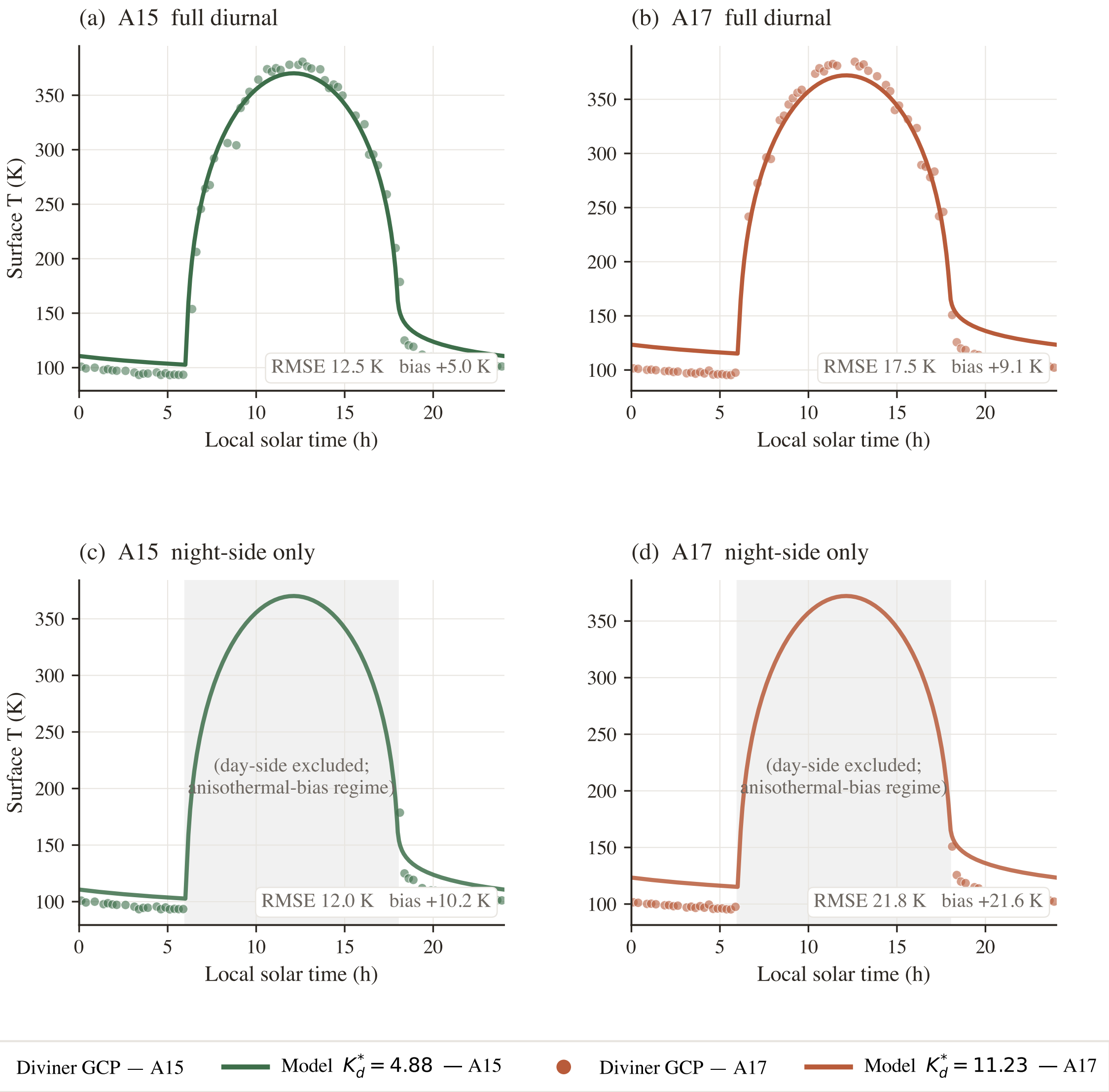

Figure 9.5. Surface-temperature closure against the Diviner GCP. (a)–(b) Full diurnal cycle at each site: dots are the Diviner brightness temperature binned at 0.25-h LST resolution within ±0.5° latitude and ±1° longitude of each landing site; solid curves are the modelled surface temperature run with the per-site retrieved K_d^*. (c)–(d) Night-side-only restriction (LST in $[0, 6) \cup [18, 24]$; day-side shaded). The full-cycle bias has a daytime anisothermal component plus a residual night-side offset that is most plausibly caused by local-scale bolometric-albedo / emissivity variations. Neither component propagates into the deep K_d retrieval, which constrains the deep equilibrium gradient Q_b/K_d and not the absolute surface temperature.

9.6 Held-out validation diagnostics

The single-parameter K_d retrieval reported in the main letter is supported by two held-out diagnostics (Fig. 9.6). *Panel (a)* reports the TG vs. TR cross-prediction: at each site we fit K_d using only the gradient-bridge (TG) sensors and evaluate the deep RMSE on the withheld ring-bridge (TR) sensors, and *vice versa*. The out-of-sample RMSEs match their in-sample counterparts to within ~ 0.2 K, indicating that the two sensor populations are mutually consistent under the same K_d retrieval. *Panel (b)* reports a leave-one-out test in which the deepest sensor at each site ($z = 139$ cm at A15, $z = 234$ cm at A17) is withheld from the fit and predicted by the remaining sensors. The model recovers the withheld temperature to within $|\Delta| = 0.11$ K (A15) and 0.16 K (A17). Both diagnostics show that the retrieval has out-of-sample skill, not just in-sample goodness-of-fit.

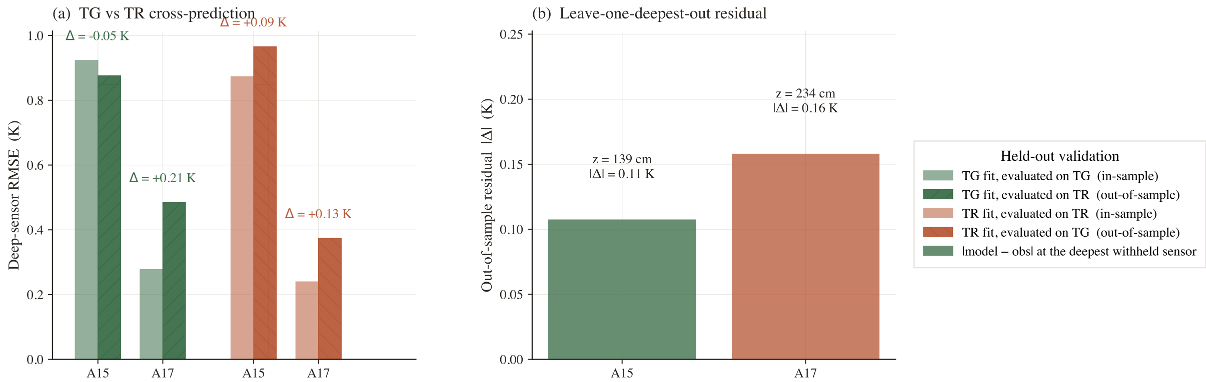

Figure 9.6. Held-out validation of the per-site K_d retrieval. (a) TG vs. TR cross-prediction: open bars show the in-sample deep RMSE achieved by the K_d^* retrieval that uses only the labelled sensor type; hatched bars show the out-of-sample RMSE on the withheld sensor type. The Δ annotations are the change between in-sample and out-of-sample RMSE. (b) Leave-one-out test on the deepest sensor at each site: the bar height is the absolute residual at the withheld sensor; the deepest-sensor depth is annotated.

Chapter 10

Glossary

ALSEP *Apollo Lunar Surface Experiments Package* — the surface-mounted electronics box that powered and recorded each Apollo geophysical experiment, including the HFE.

Bond albedo A The fraction of incident solar irradiance that a surface element scatters back to space integrated over all reflection angles and over the solar spectrum. For lunar regolith $A \sim 0.07$–0.14 depending on terrain type.

Borestem The fibreglass tube left in the HFE drill holes to keep them open. Source of the heat-short artefact discussed throughout this document.

Crank-Nicolson An unconditionally stable, second-order-accurate time-discretisation scheme for parabolic PDEs (Press et al. 2007, Chap. 20).

DE440 JPL's current planetary ephemeris, providing positions of the Sun, Moon and planets accurate to ~ 2 m over modern times.

Diviner The Diviner Lunar Radiometer Experiment on the Lunar Reconnaissance Orbiter (Paige et al., 2010) — 9 broad-band channels between 0.3 and 400 µm measuring lunar brightness temperatures.

H-parameter (H) The exponential length-scale of the depth dependence of $K(z)$ and $\rho(z)$ in the Hayne (2017) model; $H = 0.06$ m.

HFE *Heat-Flow Experiment*, the Apollo 15 and 17 instrument that drilled boreholes and lowered platinum-resistance thermometers to measure the lunar geothermal heat flux.

LST *Local solar time*, the angular position of the Sun in the local sky, measured in lunar hours from local noon.

MOON_ME The body-fixed mean Earth/principal-axis frame for the Moon used by the SPICE toolkit; the standard reference frame for lunar-surface coordinates.

Skin depth δ The depth at which a periodic surface temperature wave attenuates by e^{-1}, defined in (2.2). For the lunar diurnal wave, $\delta \sim 6$ cm.

SPICE NAIF's planetary geometry and ephemeris toolkit (Acton, 1996; Acton et al., 2018); the mission-grade source for $\theta_\odot$.

TG bridge HFE *thermal gradient* bridge: measures the temperature *difference* between two sensors ~ 50 cm apart along the borestem.

TR bridge HFE *thermometer ring* bridge: measures the *absolute* temperature of a single sensor against a reference resistor in the ALSEP electronics.

Thermal inertia I $\sqrt{K\rho c_p}$, defined in (2.1); the proportionality between surface temperature swing and inverse square-root of forcing frequency.

Bibliography

Charles Acton, Nathaniel Bachman, Boris Semenov, and Edward Wright. A look toward the future in the handling of space science mission geometry. *Planetary and Space Science*, 150: 9–12, 2018. doi: 10.1016/j.pss.2017.02.013. SPICE Toolkit; JPL DE440 ephemeris used here.

Charles H. Acton. Ancillary data services of NASA's Navigation and Ancillary Information Facility. *Planetary and Space Science*, 44(1):65–70, 1996. doi: 10.1016/0032-0633(95)00107-7.

J. L. Bandfield, P. O. Hayne, J.-P. Williams, B. T. Greenhagen, and D. A. Paige. Lunar surface roughness derived from LRO Diviner Radiometer observations. *Icarus*, 248:357–372, 2015. doi: 10.1016/j.icarus.2014.11.009.

J. Biele, E. Kuehrt, H. Senshu, N. Sakatani, K. Ogawa, M. Hamm, and M. Grott. Effects of dust layers on thermal emission from airless bodies. *Progress in Earth and Planetary Science*, 9:1–22, 2022. doi: 10.1186/s40645-022-00509-z.

I. A. Crawford. Lunar resources: A review. *Progress in Physical Geography*, 39(2):137–167, 2015. doi: 10.1177/0309133314567585.

C. J. Cremers and R. C. Birkebak. Thermal conductivity of fines from Apollo 11. *Proc. Lunar Sci. Conf.*, 2:2311–2315, 1971.

D. Foreman-Mackey, D. W. Hogg, D. Lang, and J. Goodman. emcee: The MCMC Hammer. *Publications of the Astronomical Society of the Pacific*, 125(925):306–312, 2013. doi: 10.1086/670067.

P. O. Hayne, J. L. Bandfield, M. A. Siegler, A. R. Vasavada, R. R. Ghent, J.-P. Williams, B. T. Greenhagen, O. Aharonson, C. M. Elder, P. G. Lucey, and D. A. Paige. Global Regolith Thermophysical Properties of the Moon From the Diviner Lunar Radiometer Experiment. *Journal of Geophysical Research: Planets*, 122(12):2371–2400, 2017. doi: 10.1002/2017JE005387.

B. S. Hemingway, R. A. Robie, and W. H. Wilson. Specific heats of lunar soils, basalt, and breccias from the Apollo 14, 15, and 16 landing sites between 90 and 350 K. Open-File Report 73-149, U.S. Geological Survey, 1973. Polynomial fit subsequently adopted by Hayne et al. (2017) Appendix A.

K. Horai. Thermal conductivity of lunar mare basalts at high temperature. *Proc. Lunar Sci. Conf. 12*, pages 1601–1607, 1981.

G. Kopp and J. L. Lean. A new, lower value of total solar irradiance: Evidence and climate significance. *Geophys. Res. Lett.*, 38:L01706, 2011. doi: 10.1029/2010GL045777.

R. L. Korotev. Lunar geochemistry as told by lunar meteorites. *Chemie der Erde / Geochemistry*, 65(4):297–346, 2005. doi: 10.1016/j.chemer.2005.07.001.

M. G. Langseth, S. J. Keihm, and K. Peters. Revised lunar heat-flow values. *Proc. Lunar Sci. Conf. 7*, pages 3143–3171, 1976.

A. Martínez and M. A. Siegler. A global thermal conductivity model for the surface of the Moon. *Journal of Geophysical Research: Planets*, 126(11):e2021JE006829, 2021. doi: 10.1029/2021JE006829.

E. Mazarico, G. A. Neumann, D. E. Smith, M. T. Zuber, and M. H. Torrence. Illumination conditions of the lunar polar regions using LOLA topography. *Icarus*, 211:1066–1081, 2011. doi: 10.1016/j.icarus.2010.10.030.

S. Nagihara, W. S. Kiefer, P. T. Taylor, D. R. Williams, and Y. Nakamura. Examination of the long-term subsurface warming observed at the Apollo 15 and 17 sites utilizing the newly restored heat flow experiment data from 1975 to 1977. *Journal of Geophysical Research: Planets*, 123(5): 1125–1139, 2018. doi: 10.1029/2018JE005579.

D. A. Paige, M. C. Foote, B. T. Greenhagen, J. T. Schofield, S. Calcutt, A. R. Vasavada, et al. The lunar reconnaissance orbiter Diviner lunar radiometer experiment. *Space Science Reviews*, 150:125–160, 2010. doi: 10.1007/s11214-009-9529-2.

W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery. *Numerical Recipes: The Art of Scientific Computing*. Cambridge University Press, 3rd edition, 2007.

Y. Saito, S. Tanaka, J. Takita, K. Horai, and A. Hagermann. Lunar heat flow experiment for SELENE-2. *Adv. Space Res.*, 40:1601–1607, 2007. doi: 10.1016/j.asr.2007.07.007.

N. Schorghofer and O. Aharonson. Stability and exchange of subsurface ice on Mars. *Journal of Geophysical Research: Planets*, 110(E5):E05003, 2005. doi: 10.1029/2004JE002350. The cold-trap depth framework adapted for the Moon in this work.

A. R. Vasavada, J. L. Bandfield, B. T. Greenhagen, P. O. Hayne, M. A. Siegler, J.-P. Williams, and D. A. Paige. Lunar equatorial surface temperatures and regolith properties from the Diviner Lunar Radiometer Experiment. *Journal of Geophysical Research: Planets*, 117(E12), 2012. doi: 10.1029/2011JE003987.

M. A. Wieczorek, B. L. Jolliff, A. Khan, M. E. Pritchard, B. P. Weiss, J. G. Williams, L. L. Hood, K. Righter, C. R. Neal, C. K. Shearer, I. S. McCallum, S. Tompkins, B. R. Hawke, C. Peterson, J. J. Gillis, and B. Bussey. The constitution and structure of the lunar interior. In *Reviews in Mineralogy and Geochemistry*, volume 60, pages 221–364. 2006. doi: 10.2138/rmg.2006.60.3.

J.-P. Williams, D. A. Paige, B. T. Greenhagen, and E. Sefton-Nash. The global surface temperatures of the Moon as measured by the Diviner Lunar Radiometer Experiment. *Icarus*, 283:300–325, 2017. doi: 10.1016/j.icarus.2016.08.012. GCP global cumulative product; LRO-L-DLRE-5-GCP-V1.0, DOI 10.17189/1520650.